Alimentacion 12V
J2
PWR_FLAG
PWR_FLAG
+12V
100uF
C7
C11
100n
U2
KA78M05_TO252
VI
GND
VO
V5
C15
100n
100uF
C16
Straitght key
J4
GND
Keyboard
V5
J3
GND
Paddle key
J5
DIT
DAH
GND
V5
R4
10K
V5
R5
10K
10n
C3
10n
C5
Audio input
J12
GND
T1
Transformer_1P_1S
RV1
10K
C12
100n
100R
R8
100n
C14
AUDIO_IN
V5
R11
10K
100n
C38
R12
10K
PULSADORES
SW2
Frequency
S1
S2
A
C
B
MP
MP
10n
C6
10n
C8
1n
C1
V5
JP7
PWR_FLAG
C10
100n
100n
C13
SHIELD
GND
D+
D-
5V
USB-A
J1
C2
100n
22p
C4
Y1
12MHz
C9
22p
U1
CH340T
R232
V3
VCC
UD+
UD-
XI
XO
TXD
RXD
CTS
DSR
RI
DCD
DTR
RTS
CKOH
CKOL
NOS
GND
PWR_FLAG
JP1
USB A
100n
C17
V5
10K
R14
S7
Reset
V5
C37
100n
R1
10K
R2
1K
R3
1K
R6
1K
R7
1K
R9
1K
R10
1K
R13
1K
S1
S2
S3
S4
S5
S6
S8
Mounting holes
U3
MEGA_PRO_EMBED_CH340G_slash_ATMEGA2560
AREF
RESET
RST_SPI
MOSI_SPI
MISO_SPI
SCK_SPI
5V_SPI
RX_(D0)
TX_(D1)
D2_PWM_INT-4
D3_PWM_INT-5
D4_PWM
D5_PWM
D6_PWM
D7_PWM
D8
D9_PWM
D10_PWM
D11_PWM
D12_PWM
D13_PWM
D14_(TX3)
D15_(RX3)
D16_(TX2)
D17_(RX2)
D18_(TX1)_INT-3
D19_(RX1)_INT-2
D20_INT-1
D21_INT-0
D22
D23
D24
D25
D26
D27
D28
D29
D30
D31
5V
5V_3
5V_2
3V3
3V3_1
3V3_2
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
D32
D33
D34
D35
D36
D37
D38
D39
D40
D41
D42
D43
D44-PWM
D45-PWM
D46_PWM
D47
D48
D49
D50
D51
D52
D53
VIN
VIN
GND
GND
GND
GND_1
GND_2
GND_3
1n
C18
3V3_1
AUDIO_IN
PULSADORES
DAH
DIT
Encoder
Encoder
KEY_1
KEY_2
PTT_1
PTT_2
OPT1
OPT2
AUDIO_OUT
LED1
LED2
LED3
LED4
VIN_1
VIN_2
+12V
10uF
C19
C20
100n
J6
I2C
SDA
SCL
R15
10K
R16
10K
JP3
JP2
V5
L1
10uH
Display 20 x 4
PWR_FLAG
Contrast
C21
10uF
C22
100n
DS1
LCD-20X4B
VDD
VO
E
RS
R/W
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
A
K
VSS
SHIELD
SHIELD
SHIELD
SHIELD
RV2
20K
RV3
500R
Backlight
PWR_FLAG
JP4
V5
JP5
100n
C27
R25
100K
R27
560R
R29
10R
R30
100n
C32
2N7000
Q1
100n
C28
C31
100n
3K3
AUDIO_OUT
10K
R26
2K2
R28
3K3
Q6
2N3904
Volume
C34
10uF
20K
RV4
10uF
C33
RV5
Tone
100n
C29
C30
100n
150K
R31
Q2
2N3904
C35
100n
C36
10uF
100R
R32
LS1
Speaker
J7
J8
Audio Output
J9
GND
PWR_FLAG
JP6
Side tone audio oscillator
J13
PTT_1
KEY_1
R17
220R
R19
220R
U4
PC817
U5
PC817
10n
C23
10n
C25
J10
GND
Keyer Output 1
J14
PTT_2
KEY_2
R18
220R
R20
220R
U6
PC817
U7
PC817
10n
C24
10n
C26
J11
GND
Keyer Output 2
LED4
R21
220R
D1
CW decoder indicator
LED3
R22
220R
D2
Send practice wrong
LED2
R23
220R
D3
Send practice correct
LED1
R24
220R
D4
Command mode indicator
OPT1
R33
220R
D5
Option 1
OPT2
R34
220R
D6
Option 2
Before to send the PCB to build, shortcircuit JP1, JP4, JP5 and JP6
Based on K3NG keyer and K5BCQ
By Pere López EA3AGK
Sheet: /
File: Keyer_MEGA2560.kicad_sch
Title: CW KEYER K3NG WITH MEGA2560 PRO MINI
Size: A3
Date: 2023-05-04
Rev: 1.0
KiCad E.D.A. 8.0.7
Id: 1/1